

❏ OSI Model – Easy MCQs

1.

How many layers are there in the OSI model?

- A) 5
- B) 6
- C) 7
- D) 8

Answer: C

Explanation: The OSI model consists of 7 layers.

2.

Which layer of the OSI model is responsible for physical transmission of data?

- A) Network Layer
- B) Physical Layer
- C) Transport Layer
- D) Application Layer

Answer: B

Explanation: The Physical Layer transmits raw bits over the physical medium.

3.

The Transport Layer provides which type of communication?

- A) One-way only
- B) End-to-end
- C) Between devices on the same network only
- D) Broadcast communication

Answer: B

Explanation: Transport Layer manages end-to-end communication between hosts.

4.

Which OSI layer is responsible for routing packets?

- A) Network Layer
- B) Data Link Layer
- C) Session Layer
- D) Presentation Layer

Answer: A

Explanation: The Network Layer handles routing and logical addressing.

5.

Which layer handles encryption and data formatting?

- A) Presentation Layer
- B) Transport Layer
- C) Session Layer
- D) Data Link Layer

Answer: A

Explanation: The Presentation Layer manages encryption, decryption, and data formatting.

6.

Which device operates primarily at the Data Link Layer?

- A) Router
- B) Switch
- C) Hub
- D) Firewall

Answer: B

Explanation: Switches forward frames based on MAC addresses at the Data Link Layer.

7.

What does the Application Layer provide?

- A) Hardware transmission

- B) User interface and network services
- C) Logical addressing
- D) Packet routing

Answer: B

Explanation: The Application Layer provides services for applications like email and web browsing.

8.

Which layer establishes and terminates connections between applications?

- A) Session Layer
- B) Transport Layer
- C) Network Layer
- D) Physical Layer

Answer: A

Explanation: The Session Layer manages sessions between applications.

9.

Which OSI layer is responsible for error detection in frames?

- A) Physical Layer
- B) Network Layer
- C) Data Link Layer
- D) Transport Layer

Answer: C

Explanation: The Data Link Layer performs error detection using techniques like CRC.

10.

At which OSI layer does TCP operate?

- A) Network Layer
- B) Transport Layer
- C) Session Layer
- D) Application Layer

Answer: B

Explanation: TCP is a Transport Layer protocol providing reliable communication.

☒ OSI Model – Medium MCQs

1.

Which OSI layer provides segmentation and reassembly of data?

- A) Network Layer
- B) Transport Layer
- C) Data Link Layer
- D) Application Layer

Answer: B

Explanation: The Transport Layer divides data into segments and reassembles them.

2.

Which sublayer of the Data Link Layer controls access to the physical medium?

- A) LLC (Logical Link Control)
- B) MAC (Media Access Control)
- C) Network Sublayer
- D) Session Sublayer

Answer: B

Explanation: The MAC sublayer manages access to the physical medium and MAC addressing.

3.

What is the function of the LLC sublayer in the Data Link Layer?

- A) Error detection and correction
- B) Physical addressing
- C) Multiplexing and flow control
- D) Routing packets

Answer: C

Explanation: LLC provides multiplexing, flow control, and error management services.

4.

Which OSI layer can be responsible for data compression?

- A) Presentation Layer
- B) Network Layer
- C) Transport Layer
- D) Application Layer

Answer: A

Explanation: The Presentation Layer handles data compression.

5.

What protocol does the Network Layer use for error reporting and diagnostics?

- A) ARP
- B) ICMP
- C) TCP
- D) FTP

Answer: B

Explanation: ICMP is used for error messages and operational information in the Network Layer.

6.

What is the main difference between TCP and UDP at the Transport Layer?

- A) TCP is connectionless; UDP is connection-oriented
- B) TCP provides reliable communication; UDP does not
- C) TCP operates at the Network Layer; UDP at the Transport Layer
- D) TCP is faster than UDP

Answer: B

Explanation: TCP is connection-oriented and reliable, UDP is connectionless and unreliable.

7.

Which OSI layer deals with logical addressing such as IP addresses?

- A) Transport Layer
- B) Network Layer
- C) Data Link Layer
- D) Session Layer

Answer: B

Explanation: The Network Layer assigns logical addresses (IP addresses).

8.

Which layer is responsible for determining the best physical path for data to travel?

- A) Data Link Layer
- B) Network Layer
- C) Transport Layer
- D) Application Layer

Answer: B

Explanation: The Network Layer performs routing and path determination.

9.

Which OSI layer is responsible for flow control and error control?

- A) Data Link Layer
- B) Transport Layer
- C) Network Layer
- D) Physical Layer

Answer: B

Explanation: The Transport Layer manages flow and error control end-to-end.

10.

Which protocol is used to translate IP addresses to MAC addresses?

- A) DNS
- B) DHCP
- C) ARP
- D) ICMP

Answer: C

Explanation: ARP resolves IP addresses into MAC addresses at the Data Link Layer.

11.

Which layer handles dialog control and token management?

- A) Transport Layer
- B) Network Layer
- C) Session Layer
- D) Presentation Layer

Answer: C

Explanation: The Session Layer manages dialog control and tokens for communication.

12.

At which layer does encapsulation of data into frames occur?

- A) Physical Layer
- B) Data Link Layer
- C) Network Layer
- D) Transport Layer

Answer: B

Explanation: Data Link Layer encapsulates packets into frames for transmission.

13.

Which OSI layer provides services to ensure complete data transfer?

- A) Network Layer
- B) Session Layer
- C) Transport Layer

D) Application Layer

Answer: C

Explanation: Transport Layer provides reliable data transfer services.

14.

Which OSI layer's protocol can be connectionless or connection-oriented?

A) Network Layer

B) Data Link Layer

C) Transport Layer

D) Session Layer

Answer: C

Explanation: Transport Layer supports both connectionless (UDP) and connection-oriented (TCP) protocols.

15.

Which device operates primarily at the Network Layer?

A) Switch

B) Router

C) Hub

D) Repeater

Answer: B

Explanation: Routers route packets between different networks at Layer 3.

16.

Which OSI layer is responsible for the conversion of character encoding formats?

A) Application Layer

B) Presentation Layer

C) Session Layer

D) Transport Layer

Answer: B

Explanation: The Presentation Layer converts between character encoding formats.

17.

Which layer uses MAC addresses for communication?

- A) Network Layer
- B) Data Link Layer
- C) Transport Layer
- D) Physical Layer

Answer: B

Explanation: MAC addresses are used at the Data Link Layer for local network communication.

18.

In the OSI model, which layer directly interacts with software applications?

- A) Transport Layer
- B) Application Layer
- C) Session Layer
- D) Presentation Layer

Answer: B

Explanation: The Application Layer provides services directly to software applications.

19.

What is the unit of data at the Network Layer?

- A) Bit
- B) Frame
- C) Packet
- D) Segment

Answer: C

Explanation: The Network Layer handles data packets.

20.

Which OSI layer deals with synchronization of data streams?

- A) Session Layer
- B) Presentation Layer
- C) Transport Layer
- D) Network Layer

Answer: A

Explanation: Session Layer synchronizes data streams between communicating systems.

21.

Which OSI layer is concerned with hardware addressing?

- A) Physical Layer
- B) Data Link Layer
- C) Network Layer
- D) Transport Layer

Answer: B

Explanation: Data Link Layer manages hardware (MAC) addressing.

22.

Which of the following is NOT a function of the Transport Layer?

- A) Flow control
- B) Routing
- C) Segmentation
- D) Error recovery

Answer: B

Explanation: Routing is done by the Network Layer, not the Transport Layer.

23.

Which layer is responsible for defining protocols like HTTP and FTP?

- A) Transport Layer
- B) Application Layer
- C) Presentation Layer

D) Session Layer

Answer: B

Explanation: Application Layer protocols include HTTP, FTP, SMTP, etc.

24.

Which OSI layer uses port numbers to identify applications?

A) Session Layer

B) Transport Layer

C) Network Layer

D) Data Link Layer

Answer: B

Explanation: Transport Layer uses port numbers to differentiate applications.

25.

The process of encapsulation adds headers at which layers?

A) All OSI layers except Physical

B) Only Data Link and Network

C) Only Network and Transport

D) Only Application and Presentation

Answer: A

Explanation: Encapsulation involves adding headers at each layer except Physical.

26.

Which layer is responsible for framing?

A) Data Link Layer

B) Network Layer

C) Transport Layer

D) Session Layer

Answer: A

Explanation: Framing happens at the Data Link Layer.

27.

Which OSI layer typically uses IP addresses?

- A) Network Layer
- B) Data Link Layer
- C) Transport Layer
- D) Application Layer

Answer: A

Explanation: IP addressing is a Network Layer function.

28.

In which OSI layer does SSL/TLS operate?

- A) Session Layer
- B) Presentation Layer
- C) Transport Layer
- D) Application Layer

Answer: B

Explanation: SSL/TLS operates at the Presentation Layer providing encryption.

29.

Which OSI layer manages the establishment of virtual circuits?

- A) Network Layer
- B) Data Link Layer
- C) Session Layer
- D) Transport Layer

Answer: C

Explanation: Session Layer manages virtual circuit establishment and termination.

30.

Which OSI layer converts data into a form the application can use?

- A) Application Layer
- B) Presentation Layer
- C) Session Layer
- D) Transport Layer

Answer: B

Explanation: The Presentation Layer translates data between network and application formats

1.

Which of the following statements about the OSI Transport Layer is TRUE?

- A) It guarantees packet delivery without error by retransmitting lost packets using UDP.
- B) It handles segmentation but does not provide multiplexing of data streams.
- C) It manages end-to-end connections and can provide both reliable and unreliable delivery services.
- D) It is responsible for physical addressing and bit-level transmission.

Answer: C

Explanation: The Transport Layer manages end-to-end connections and can provide reliable (TCP) or unreliable (UDP) delivery.

2.

Which OSI layer can be involved in the negotiation of communication parameters such as encryption algorithms and data compression before actual data transfer?

- A) Session Layer
- B) Presentation Layer
- C) Application Layer
- D) Network Layer

Answer: B

Explanation: The Presentation Layer negotiates data representation such as encryption and compression.

3.

Which protocol or technology operates across multiple OSI layers, making strict layer assignment difficult?

- A) ARP
- B) MPLS (Multi-Protocol Label Switching)

- C) ICMP
- D) HTTP

Answer: B

Explanation: MPLS operates at a layer often considered between Layer 2 and Layer 3, making strict classification challenging.

4.

In an OSI-based network, which layer is responsible for maintaining session checkpoints and recovery after a disruption?

- A) Transport Layer
- B) Session Layer
- C) Network Layer
- D) Presentation Layer

Answer: B

Explanation: The Session Layer maintains checkpoints and recovery mechanisms for ongoing sessions.

5.

Consider a data packet encapsulated in an Ethernet frame traveling from host A to host B across multiple routers. Which OSI layer's header remains unchanged throughout the journey?

- A) Data Link Layer header
- B) Network Layer header
- C) Transport Layer header
- D) Physical Layer header

Answer: C

Explanation: Transport Layer headers (like TCP/UDP) remain unchanged; Data Link Layer headers are rewritten at each router.

6.

Which OSI layer is most directly responsible for performing fragmentation and reassembly of packets?

- A) Data Link Layer
- B) Network Layer
- C) Transport Layer
- D) Session Layer

Answer: B

Explanation: The Network Layer fragments and reassembles packets to accommodate different MTUs.

7.

Which OSI layer uses tokens to prevent collision and ensure orderly communication in a multi-access network?

- A) Data Link Layer (MAC sublayer)
- B) Network Layer
- C) Transport Layer
- D) Session Layer

Answer: A

Explanation: Token-based access methods (like Token Ring) operate at the MAC sublayer of the Data Link Layer.

8.

In the OSI model, which layer's protocols provide flow control mechanisms to prevent overwhelming the receiver?

- A) Network Layer and Data Link Layer
- B) Transport Layer and Data Link Layer
- C) Session Layer and Presentation Layer
- D) Application Layer and Transport Layer

Answer: B

Explanation: Flow control occurs at both Data Link (e.g., sliding window) and Transport Layers (TCP flow control).

9.

Which OSI layer protocol is responsible for multiplexing several communication streams over a single physical channel?

- A) Presentation Layer
- B) Session Layer
- C) Transport Layer
- D) Data Link Layer

Answer: D

Explanation: The Data Link Layer multiplexes multiple logical links over a physical medium, using techniques like VLAN tagging.

10.

Which OSI layer protocol can be associated with both connection-oriented and connectionless communication depending on implementation?

- A) Network Layer
- B) Transport Layer
- C) Session Layer
- D) Application Layer

Answer: A

Explanation: The Network Layer can be connectionless (IP) or connection-oriented (X.25, ATM) depending on the protocol.

That completes the OSI Model MCQs set!